

Notes: Brunnermeier, Sockin, and Xiong (RES, 2022)

China's Model of Managing the Financial System

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1 Big picture

The paper builds a stylized asset-market model to formalize a distinctive *policy trade-off* that is especially salient in China:

- **Financial stability channel:** if the government trades against destabilizing (noise) demand, it can dampen price volatility and prevent “volatility explosions”.
- **Price efficiency channel:** stabilizing interventions create *policy noise* (imperfect implementation / discretionary policy shocks). Investors then rationally allocate their scarce information capacity to predict policy noise rather than fundamentals, so prices become *less informative* about fundamentals.

A key message is that **stability vs. efficiency is endogenous**: aggressive stabilization can shift the market into a *government-centric* equilibrium where private information acquisition and trading focus on policy.

2 Model map

There are four nested “models” in the paper:

1. **Baseline (perfect information) asset market** with risk-averse investors and noise traders $\Rightarrow$ potential *volatility explosion*.
2. **Government intervention** as a trading program that leans against noise demand but adds *policy noise* that scales with intensity.
3. **Information frictions + endogenous information choice** where investors can acquire *either* a signal about fundamentals or a signal about policy noise $\Rightarrow$ multiple equilibria (fundamental-centric vs. government-centric).
4. **Micro-founded welfare and optimal intervention** where the government chooses intensity taking into account the induced equilibrium information regime.

3 Baseline asset-market model (perfect information)

3.1 Cash flows and fundamentals

There is a single risky asset with cash flow (dividend)

$$D_t = V_t + \sigma_D \varepsilon_t^D, \quad \varepsilon_t^D \sim \mathcal{N}(0, 1), \quad (1)$$

$$V_t = \rho_V V_{t-1} + \sigma_V \varepsilon_t^V, \quad \varepsilon_t^V \sim \mathcal{N}(0, 1), \quad (2)$$

with $|\rho_V| < 1$. The gross risk-free rate is constant: $R^f > 1$.

Let P_t be the time- t price, and define the one-period excess payoff:

$$R_{t+1} \equiv D_{t+1} + P_{t+1} - R^f P_t. \quad (3)$$

3.2 Agents: investors and noise traders

Risk-averse investors. A continuum of competitive investors (mass 1) have CARA utility over next-period wealth and behave myopically (mean–variance):

$$U_t^i = \mathbb{E}[-\exp(-\gamma W_{t+1}^i) \mid \mathcal{F}_t], \quad W_{t+1}^i = R^f \bar{W} + X_t^i R_{t+1}, \quad (4)$$

where X_t^i is the investor’s position in the risky asset and $\gamma > 0$ is risk aversion.

Noise traders. Noise traders submit exogenous net demand

$$N_t = \sigma_N \varepsilon_t^N, \quad \varepsilon_t^N \sim \mathcal{N}(0, 1). \quad (5)$$

Market clearing. The risky asset is in zero net supply:

$$\int_0^1 X_t^i di + N_t = 0. \quad (6)$$

Under perfect information, V_{t+1} and N_t are observed by all investors at date t .

3.3 Linear equilibrium and “volatility explosion”

The baseline equilibrium is linear in V_{t+1} and N_t :

$$P_t = \frac{1}{R^f - \rho_V} V_{t+1} + p_N N_t. \quad (7)$$

Investor demand. Given CARA–normality, the optimal demand is

$$X_t^i = \frac{1}{\gamma} \frac{\mathbb{E}[R_{t+1} \mid \mathcal{F}_t]}{\text{Var}(R_{t+1} \mid \mathcal{F}_t)}. \quad (8)$$

In this equilibrium, the expected excess payoff is driven by mean reversion of the noise-trader-induced mispricing:

$$\mathbb{E}[R_{t+1} \mid \mathcal{F}_t] = -R^f p_N N_t. \quad (9)$$

The conditional return variance is

$$\text{Var}(R_{t+1} \mid \mathcal{F}_t) = \underbrace{\sigma_D^2 + \left(\frac{\sigma_V}{R^f - \rho_V} \right)^2}_{\text{cash-flow \& news risk}} + \underbrace{p_N^2 \sigma_N^2}_{\text{next-period noise-trader risk}}. \quad (10)$$

Hence

$$X_t^i = -\frac{1}{\gamma} \frac{R^f p_N}{\sigma_D^2 + \left(\frac{\sigma_V}{R^f - \rho_V} \right)^2 + p_N^2 \sigma_N^2} N_t. \quad (11)$$

Solving for p_N and the breakdown condition. Market clearing (6) implies $X_t^i = -N_t$. Let $\bar{\sigma}^2 \equiv \sigma_D^2 + \left(\frac{\sigma_V}{R^f - \rho_V} \right)^2$. Then p_N solves

$$\frac{\gamma \sigma_N^2}{R^f} p_N^2 - p_N + \frac{\gamma \bar{\sigma}^2}{R^f} = 0. \quad (12)$$

A real solution exists iff the discriminant is nonnegative:

$$\sigma_N \leq \sigma_N^* \equiv \frac{R^f}{2\gamma\sqrt{\bar{\sigma}^2}}. \quad (13)$$

Interpretation (Proposition 1). As $\sigma_N \uparrow \sigma_N^*$, investors must absorb more noise-trader risk but have finite risk-bearing capacity; the feedback through p_N causes *return volatility to explode*. For $\sigma_N > \sigma_N^*$, no stationary linear RE equilibrium exists.

4 Government intervention as trading (stability vs. policy noise)

4.1 Intervention rule

The government acts as a large stabilizing trader that leans against noise-trader demand but introduces implementation noise:

$$X_t^G = -\vartheta_N N_t + \sigma_N \vartheta_N G_t, \quad G_t \sim \mathcal{N}(0, \sigma_G^2). \quad (14)$$

Here $\vartheta_N \geq 0$ is intervention intensity:

- $-\vartheta_N N_t$: “leaning against the wind”,
- $\sigma_N \vartheta_N G_t$: policy noise that grows with intensity.

Market clearing becomes $\int X_t^i di + N_t + X_t^G = 0$. From the investors’ viewpoint, the “effective” non-fundamental order flow is

$$(1 - \vartheta_N)N_t + \sigma_N \vartheta_N G_t.$$

4.2 Linear equilibrium with intervention

The price remains linear:

$$P_t = \frac{1}{R^f - \rho_V} V_{t+1} + p_N N_t + p_g G_t. \quad (15)$$

4.3 Main comparative statics (Proposition 2)

Key object: the variance of effective order flow. Because N_t and G_t are independent,

$$\text{Var}((1 - \vartheta_N)N_t + \sigma_N \vartheta_N G_t) = \sigma_N^2 \left[(1 - \vartheta_N)^2 + \vartheta_N^2 \sigma_G^2 \right]. \quad (16)$$

Interpretation of Proposition 2 (in words):

- For *moderate* ϑ_N (policy noise not too large relative to stabilization), intervention **reduces return volatility** and **raises price informativeness**.
- The “volatility explosion” threshold shifts outward because the effective noise load is scaled by $\sqrt{(1 - \vartheta_N)^2 + \vartheta_N^2 \sigma_G^2}$.
- For *aggressive* ϑ_N , policy noise dominates and intervention can **increase** volatility (the government becomes the main non-fundamental shock).

5 Information frictions and endogenous information choice

5.1 Why add information frictions?

With perfect information, the government can stabilize without affecting what investors know. The paper's core "China" mechanism requires:

1. **imperfect information about fundamentals and noise trading**, so prices play an informational role;
2. **policy noise**, so policy itself becomes a priced risk factor;
3. **endogenous attention**, so investors can choose to learn fundamentals or policy.

5.2 Public information and belief objects

Public information at date t is

$$\mathcal{F}_t^M = \{D_s, P_s, G_s\}_{s \leq t}.$$

Key public beliefs:

$$\hat{V}_{t+1}^M \equiv \mathbb{E}[V_{t+1} \mid \mathcal{F}_t^M], \quad \hat{N}_t^M \equiv \mathbb{E}[N_t \mid \mathcal{F}_t^M], \quad \hat{G}_{t+1}^M \equiv \mathbb{E}[G_{t+1} \mid \mathcal{F}_t^M].$$

5.3 Government intervention with filtering

The government leans against the *market's* estimate $\hat{N}_t^M$ and adds noise proportional to the conditional variance of its action:

$$X_t^G = -\vartheta_{\hat{N}} \hat{N}_t^M + \sqrt{\text{Var}(\vartheta_{\hat{N}} \hat{N}_t^M \mid \mathcal{F}_{t-1}^M)} G_t. \quad (17)$$

5.4 Private signals and discrete information choice

Each investor chooses $a_t^i \in \{0, 1\}$:

- If $a_t^i = 1$, acquire a **fundamental** signal:

$$s_t^i = V_{t+1} + \frac{1}{\sqrt{\tau_s}} \varepsilon_t^{s,i}.$$

- If $a_t^i = 0$, acquire a **policy-noise** signal:

$$g_t^i = G_{t+1} + \frac{1}{\sqrt{\tau_g}} \varepsilon_t^{g,i}.$$

5.5 NREE price function (key linear conjecture)

In a noisy rational expectations equilibrium (NREE), the equilibrium price is linear in public beliefs and innovations:

$$P_t = \frac{1}{R^f - \rho_V} \hat{V}_{t+1}^M + p_g G_t + p_{\hat{G}} \hat{G}_{t+1}^M + p_V (V_{t+1} - \hat{V}_{t+1}^M) + p_G (G_{t+1} - \hat{G}_{t+1}^M) + p_N N_t. \quad (18)$$

5.6 Investor demand and the information-choice problem

Demand (Equation (4.4)). With CARA-normality,

$$X_t^i = \frac{1}{\gamma} \frac{\mathbb{E}[R_{t+1} \mid \mathcal{F}_t^i]}{\text{Var}(R_{t+1} \mid \mathcal{F}_t^i)}. \quad (19)$$

Information choice (Equation (4.5)). Given $\mathcal{F}_t^M$, choose a_t^i to minimize the conditional variance of R_{t+1} :

$$\max_{a_t^i \in \{0,1\}} \left\{ -\text{Var}(R_{t+1} \mid \mathcal{F}_t^M, a_t^i) \right\}. \quad (20)$$

Selection condition (Proposition 3). A signal $z \in \{s, g\}$ reduces conditional variance in proportion to

$$\frac{\text{Cov}(R_{t+1}, z_t^i \mid \mathcal{F}_t^M)^2}{\text{Var}(z_t^i \mid \mathcal{F}_t^M)}.$$

Choose the signal that yields the larger variance reduction.

5.7 Equilibrium regimes and a key intuition for volatility

Because prices aggregate information, information choice is strategic and can yield multiple equilibria:

- **Fundamental-centric:** most investors learn V_{t+1} .
- **Government-centric:** most investors learn G_{t+1} .

Why can the government-centric equilibrium have *lower* volatility? In the government-centric equilibrium, investors share (more) aligned beliefs about policy and tend to trade *in the same direction* as the government when absorbing noise-trader demand. In contrast, in a fundamental-centric equilibrium, investors have heterogeneous fundamental signals and therefore trade more against each other (and against the government), which increases trading volume and can raise return volatility even if prices are more informative.

5.8 Existence of a government-centric equilibrium (Proposition 4, i.i.d. case)

In a simplified i.i.d. case, the paper gives a necessary and sufficient condition for existence of a *government-centric* equilibrium (paper's Eq. (4.7)):

$$\begin{aligned} & \frac{1}{2\sigma_N c} \frac{R^f}{1 - \vartheta_{\hat{N}}} - \sqrt{\left(\frac{1}{2\sigma_N c} \frac{R^f}{1 - \vartheta_{\hat{N}}} \right)^2 - \frac{\sigma_V^2 + \sigma_D^2}{c}} \\ & \geq \frac{\sigma_V^2}{\sqrt{\sigma_V^2 + \tau_s^{-1}}} (1+x) \sqrt{(\sigma_G^2 + (1+x)\tau_g^{-1}) \left(\frac{\frac{1-\vartheta_{\hat{N}}}{\vartheta_{\hat{N}}}}{\sigma_G^2 - R^f \frac{x}{1-\vartheta_{\hat{N}}}} \right)^2}, \end{aligned} \quad (21)$$

where x solves $x(1+x)^3 = \left(\frac{\vartheta_{\hat{N}}}{R^f \sigma_G^3} \right)^2$ and $c \geq 0$ is a parameter function (given in the supplementary material).

6 Calibration and numerical illustrations

6.1 Table 1: baseline parameters (interpretation)

Baseline parameter values (Table 1):

Table 1: Baseline parameter values (Table 1 in the paper)

Government		Asset fundamental	
$\gamma\sigma$	1.25	ρ_V	0.75
γ_W	1	σ_V^2	0.01
σ_G^2	2	σ_D^2	0.80
Noise trading		Investors	
σ_N^2	0.20	γ	1
		τ_s	500
		τ_g	500
		R^f	1.01

Notes to myself:

- $\rho_V = 0.75$ makes fundamentals persistent, so public learning from prices is meaningful.
- $\sigma_G^2 = 2$ implies sizable policy noise when intervention is strong.
- $\tau_s = \tau_g = 500$ makes private signals quite precise, so the equilibrium *allocation of attention* matters for both volatility and informativeness.

6.2 Figure 1: intervention reduces volatility but can backfire

Figure 1 plots conditional return volatility as a function of noise-trader risk σ_N^2 :

- **No intervention:** volatility increases in σ_N^2 and becomes arbitrarily large as σ_N approaches the breakdown threshold σ_N^* .
- **With intervention:** for moderate ϑ_N , volatility is lower for given σ_N^2 and the breakdown threshold shifts outward.
- **But:** strong intervention introduces policy noise, limiting stabilization.

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6.3 Figure 2: equilibrium switch as intervention intensity rises

Figure 2 varies intervention intensity ϑ_N and plots:

- **Panel A:** conditional return volatility,
- **Panel B:** conditional variance of price deviation from the benchmark $\frac{1}{R^f - \rho_V} V_{t+1}$.

Main pattern:

1. For low ϑ_N , the equilibrium is **fundamental-centric**. Increasing ϑ_N reduces volatility and improves price efficiency.
2. Beyond a threshold, the economy switches to **government-centric**. Volatility can fall further, but *price efficiency deteriorates sharply*: mispricing variance jumps up because investors endogenously focus on policy-noise prediction.

7 Welfare and optimal intervention

7.1 Microfoundation: who is in the economy?

For welfare, the paper micro-founds three relevant constituencies:

- **Entrepreneurs/firms** who use price information to allocate capital to projects (so they value price informativeness about V).
- **Investors** who trade and can acquire private information (so they value profitable trading opportunities).
- **Liquidity traders and taxpayers** who are exposed to non-fundamental risk and bear the (stochastic) costs of policy implementation.

Liquidity traders (Appendix B intuition). A representative reduced-form for liquidity trader j 's net order is:

$$n_t^j = N_t + \sigma_n \varepsilon_t^j, \quad \varepsilon_t^j \sim \mathcal{N}(0, 1), \quad (22)$$

so the *aggregate* order flow is N_t (which moves prices), while the idiosyncratic component averages out in prices but still affects traders' risk exposure and thus welfare (via σ_n^2).

Social welfare aggregator. The government uses a Nash-style social welfare function over groups (equivalently, a monotonic transformation of the product of expected utilities). Under CARA-normality and zero-sum trading profits, welfare can be reduced to an objective that depends on *second moments* (volatility and informativeness).

7.2 Government objective (Proposition 5 / Eq. (5.8))

In the baseline specification, the government's welfare problem can be written as:

$$\begin{aligned} \sup_{\vartheta_{\hat{N}}} & \underbrace{\frac{\sigma_V^2}{(1 - \rho_V^2) \Sigma^{M, VV} + \sigma_y^2}}_{\text{(i) production efficiency via price informativeness}} - \underbrace{\frac{\text{Var}(R_{t+1} \mid \mathcal{F}_{t-2}^M)}{\text{Var}(R_{t+1} \mid \mathcal{F}_t^i)}}_{\text{(ii) investors' info advantage / trading gains}} \\ & - \underbrace{\gamma^2 \left(\sigma_N^2 + \sigma_n^2 + \vartheta_{\hat{N}}^2 (1 + \sigma_G^2) (\sigma_N^2 - \Sigma^{M, NN}) \right) \text{Var}(R_{t+1} \mid \mathcal{F}_t^M)}_{\text{(iii) risk-bearing \& policy-noise costs}}. \end{aligned} \quad (23)$$

Here $\Sigma^{M, VV}$ is the public posterior variance of V and $\Sigma^{M, NN}$ is the public posterior variance of N .

7.3 Sufficient statistics (Proposition 6)

Conditional on an equilibrium regime:

- In a **fundamental-centric** equilibrium, optimal intervention depends on return volatility $\text{Var}(R_{t+1} \mid \mathcal{F}_t^M)$ and on $\Sigma^{M,VV}$.
- In a **government-centric** equilibrium, the relevant informational object is $\Sigma^{M,GG}$ instead of $\Sigma^{M,VV}$.

7.4 Figure 3: optimal policy can induce an equilibrium switch

Figure 3 varies noise-trader risk σ_N^2 and plots outcomes under the *optimal* policy choice $\vartheta_{\hat{N}}$:

- At low σ_N^2 : the government chooses modest intervention $\Rightarrow$ **fundamental-centric**.
- At high σ_N^2 : intensive intervention becomes optimal and can induce a switch $\Rightarrow$ **government-centric**.

This switch lowers volatility but increases mispricing variance; welfare can remain smooth because the government optimally trades off these forces.

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[width=]f4.png

7.5 Figure 4: when does the “China-style” outcome arise?

Figure 4 plots the boundary between equilibria under the government’s optimal policy in the (σ_y^2, σ_N^2) plane:

- Higher σ_N^2 pushes toward **government-centric** outcomes (stability is worth more).
- Higher σ_y^2 also pushes toward **government-centric** outcomes because price informativeness about V matters less for real investment when project-level noise is high.

8 Short summary

The model rationalizes a “China-style” regime as a welfare-relevant equilibrium:

1. The government leans against destabilizing trading to maintain stability.
2. But intervention adds policy noise.
3. Investors optimally allocate attention to predicting policy, not fundamentals.
4. Hence prices can be stable yet *informationally inefficient*.

Which regime is optimal depends on underlying conditions (noise-trader risk; how valuable price signals are for real allocation).

9 Conclusion

9.1 Summary

This paper provides a unified theoretical explanation for why China and Western economies can rationally adopt very different approaches to managing financial markets. The key policy trade-off is between **financial stability** (lower price volatility and fewer destabilizing fluctuations) and **price efficiency** (prices reflecting fundamentals and guiding real allocation). In the model, the government can stabilize markets by *leaning against* noise-trader demand through intervention, but intervention inevitably introduces **policy noise** (implementation frictions and uncertainty about policy actions), which becomes an additional non-fundamental driver of prices.

The welfare analysis identifies economic conditions under which **intensive and regular intervention is optimal**. When *noise-trader risk* is sufficiently high and/or when *idiosyncratic production risk* is sufficiently large (so that fundamental price signals are less useful for real investment decisions), the welfare-maximizing policy pushes the economy into a **government-centric equilibrium**. In that equilibrium, investors optimally devote their private information acquisition to forecasting future policy noise rather than learning fundamentals. Prices can then be *stable* yet *informationally inefficient*.

Conversely, when noise-trader risk is lower and when the real economy benefits more from informative prices, the welfare-maximizing policy is **more modest intervention**, sustaining a **fundamental-centric equilibrium** in which investors learn fundamentals and prices are more informative—a regime that resembles the Western approach.

A central conclusion is therefore:

China-style intensive intervention can be welfare-optimal under certain underlying economic conditions, even without invoking political objectives, but it comes with an endogenous cost: reduced informational content of prices.

9.2 Intuition: one continuous mechanism

The mechanism is most transparent if read as one chain:

1. **Noise trading creates extrinsic volatility.** As noise-trader risk rises, prices become more volatile because investors demand compensation to absorb noise-trader positions, which amplifies volatility through an equilibrium feedback.
2. **Government intervention stabilizes by trading against noise traders.** By partially offsetting noise-trader demand, intervention can reduce return volatility and mitigate volatility “explosions.”
3. **But intervention adds an extra source of uncertainty: policy noise.** Implementation frictions mean the government’s intervention is not perfectly predictable; this injects policy-related shocks into prices.
4. **Investors respond endogenously by reallocating attention.** With information frictions, investors choose what to learn. When intervention becomes sufficiently intensive, policy noise becomes an important driver of future returns, so learning policy becomes privately valuable.
5. **A regime shift can occur: fundamentals → policy.** Beyond a threshold, investors collectively switch from acquiring fundamental information to acquiring policy-noise information.

This shift can *lower volatility* while *worsening price efficiency* (prices deviate more from fundamentals).

6. **Welfare chooses the regime, not volatility alone.** The government internalizes the volatility–efficiency trade-off. When fundamental price signals are less socially valuable (e.g., high idiosyncratic project noise), the welfare-maximizing choice can prioritize stability, even if it reduces price informativeness.

9.3 Key implications and testable predictions

The paper implies several sharp predictions that are useful for empirical work:

- **Stability does not imply informativeness.** Periods of intensive intervention can exhibit lower volatility but weaker price discovery and greater disconnect between prices and fundamentals.
- **Intervention intensity can generate a discontinuous regime change.** As policy becomes more aggressive, the market can transition from fundamental-centric to government-centric, creating a discrete change in the joint behavior of volatility and mispricing.
- **As the economy develops, the optimal regime can shift.** If prices become more important for guiding real investment (and/or noise-trader risk declines), the welfare-maximizing policy becomes less interventionist; the model predicts that loosening intervention can raise volatility while improving informational efficiency.
- **Information content shifts toward policy.** In the government-centric regime, investor learning and market commentary should tilt toward predicting government actions and policy implementation noise rather than firm-level fundamentals.

9.4 Takeaway

The “China model” can be welfare-optimal because aggressive intervention stabilizes markets facing high noise-trader risk, but it endogenously crowds investors into learning policy noise rather than fundamentals, producing stable yet informationally inefficient prices.